Syllabus

UG0802 -ZOO-64T-203- Cell Biology, Genetics& Biotechnology

UG0802 -ZOO-64P-204- Practicals based onCell Biology,Genetics& Biotechnology

IV Semester-Zoology

<table>
<tr><th>Semester</th><th>Code of the Course</th><th colspan="3">Title of the Course/Paper</th><th>NHEQF Level</th><th>Credits</th></tr>
<tr><td>IV</td><td>ZOO-64T-203
ZOO-64P-204</td><td colspan="3">Cell Biology, Genetics& Biotechnology
Practicals based onCell Biology, Genetics& Biotechnology</td><td>6</td><td>6</td></tr>
<tr><th rowspan="2">Level of Course</th><th rowspan="2">Type of the Course</th><th colspan="3">Credit Distribution</th><th rowspan="2">Offered to NC Student</th><th rowspan="2">Course Delivery Method</th></tr>
<tr><th>Theory</th><th>Practical</th><th>Total</th></tr>
<tr><td>6</td><td>Major</td><td>4</td><td>2</td><td>6</td><td>Yes</td><td>Lectures: 60 lectures including diagnostic and informative assessments during lecture hours and 30 Hours of Practical training/demonstration.</td></tr>
<tr><td colspan="2">List of Programme Codes in which Offered as Minor Discipline</td><td colspan="5">B.Sc. Chemistry: UG0804
B.Sc. Botany: UG0805</td></tr>
<tr><td colspan="2">Prerequisites</td><td colspan="5">B. Sc (Bio Group) IIISEM</td></tr>
<tr><td colspan="2">Objectives of the Course:</td><td colspan="5">This course will provide with a deep knowledge of Cell Biology, Genetics and Biotechnology.
• Understand the role of different cell organelles in the maintenance of life activities, the history and basic concepts of heredity, variations and gene interaction, the application of biotechnology in the fields of industry and agriculture.
• In addition to this, the course is aimed at nurturing skills of conducting scientific inquiry and experimentation in the field of recent advancements, recent trends and technologies.
• Students can start their own business i.e. self-employmentsand get employment in different applied sectors.</td></tr>
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Detailed syllabus

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UG0802 -ZOO-64T-203- Cell Biology, Genetics& Biotechnology

UNIT I

Introduction to cell: Morphology, size, shape, characteristics and structure of prokaryotic and eukaryotic animal cells; basic idea of virus and cell theory. **3 Hrs**

Plasma membrane: Composition, Fluid mosaic model; Transport across the membrane: active and passive transport, facilitated transport, diffusion, osmosis. **4 Hrs**

Cell Organelles: Structure and functions of nucleus, mitochondria, endoplasmic reticulum, ribosome (prokaryotic and eukaryotic), Golgi complex, lysosome, microbodies and centrioles. Structure and functions of cilia, flagella, and microvilli. **8 Hrs**

UNIT II

Cell Division: Mitosis, Meiosis, cell cycle. **4 Hrs**

Prokaryotic chromosomes and eukaryotic chromosomes: Morphology, telomeres, primary and secondary constrictions, chromatids; Giant chromosome types: Polytene and Lampbrush. **4Hrs**

Chromosomal organization: Euchromatin, heterochromatin, folded fiber model and nucleosome concept. **4 Hrs**

Cell-Cell Junctions: Structure and Functions- Tight junctions, Desmosomes, Gap junctions. **3 Hrs**

UNIT III

Mendel's law of Inheritance: Principle of segregation, independent assortment, dominance, Mendelian genetics in humans, Variety of gene expression- modifiers, suppressors, pleiotropic gene, multiple alleles, Interaction of gene epistasis, complimentary gene, duplicate gene. **4 Hrs**

Linkage: Definition, coupling and repulsion hypothesis, linkage groups, Crossing over-mechanism and theory; structure of chromosomes, extrachromosomal inheritance-mitochondrial and plastids. **4 Hrs**

Mutation: Definition, basic concept, also include types (spontaneous and induced); mechanism of mutagenesis; base analogues, nitrous acid, hydroxyl amine, alkylating agent, Acridine dyes, U.V. light. **4 Hrs**

Genetic disorders: Down's, Turner's and Klinefelter's syndromes, color blindness, Hemophilia and Phenylketonuria. **3 Hrs**

UNIT IV

Concept and scope of Animal Biotechnology: Cloning methods (Cell, Animal and Gene cloning). Vectors- Plasmids, Cosmids, Lambda bacteriophages and Yeast artificial chromosomes (YAC). **4 Hrs**

Animal cell culture: Equipment and materials for animal cell culture; applications of cell culture techniques; Recombinant DNA technology and its applications. **4 Hrs**

Transgenesis: Methods of Transgenesis, Production of transgenic animals and their application in Biotechnology; Stem cells: Types and their applications. **4Hrs**

Dairy biotechnology: Food, drink and dairy biotechnology (outline idea only). Fermented food production: dairy products, wine, beer, vinegar and food preservation. **3Hrs**

Suggested Books and References:

1. Lodish,H.,Berk,A.,Zipursky,S.L.,Matsudaira,P.,Baltimore,D.andJamesDarnell,J.MolecularCellBiology,Freeman,7th edition2013.
2. Cell Biology, Genetics, Molecular biology, Evolution and Ecology(2022) P.S. Verma, V.K.Agarwal.
3. CrossA.E. andNagleR.B.(2006).CellAdhesionandCytoskeletalMoleculesin Metastasis. Vol. XII, Springer Publication.
4. KarpG.John.(2013).CellandMolecularBiology.ConceptsandExperiments.7th Edition, Wiley & Sons Inc., New York.
5. Griffiths,A.J.F., J.H. Miller, Suzuki, D.T., Lewontin, R.C. and Gelbart, W.M. (2009). An Introduction to Genetic Analysis. IX Edition. Freeman and Co., N.Y., USA.
6. Brown,T.A.(2015).GeneCloningandDNA Analysis.7th Edition, AcademicPress, California, USA.
7. GardnerE.J.(2008).PrinciplesofGenetics.VIIIEdition,SimmonsM.J.andSnustadD.P.Willey, India.
8. PierceB.A.(2008).Genetics-AConceptualApproach.W.H.Freeman&Co.,New York.
9. Watson, J.D., Myers, R.M., Caudy,A.andWitkowski, J.K. (2007). Recombinant DNA-Genes and Genomes- A Short Course. III Edition, Freeman and Co., N.Y., USA.
10. BiotechnologybyU.Satyanarayan.(2010).
11. B.D.Singh.(2004).Biotechnology-ExpandingHorizons.KalyaniPublishers,New Delhi. India.
12. CurrentFrontiersandPerspectivesinCellBiology (2012).StevoNajman.
13. Cooper, G. M., and Hausman, R. E. (2013). The Cell: A MolecularApproach (6th Ed.).Washington: ASM; Sunderland.
14. PrinciplesofGeneticsbyGardner (2008) (8th Edition).
15. Genetics(2009)P.K.Gupta,RastogiPublications.
16. PrimroseS. B. andTwymanR. M: Principlesof Gene Manipulation and Genomics. John Wiley & Sons, 2013.

Suggested E-Resources:

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1. The Cell: A Molecular Approach (2013) Geoffrey M. Cooper and Robert E. Hausman. Sixth Edition. Sinauer Associates.
2. Principles of Molecular Biology (2023) Veer Bala Rastogi. Second Edition. Medtech.
3. Genetics and Molecular Biology (Volume 1) Kohji Hasanuma. Encyclopedia of Life Support Systems. UNESCO-EOLSS.
4. https://egyankosh.ac.in/handle/123456789/5504

Course Learning Outcome: Upon completion of the course, students will be able to:

- StudentswillbeabletoexplainthebasicconceptsofCellBiology.
- Haveanunderstandingofclassicalgenetics.
- To impart knowledge and practical training in various techniques to develop research in commercial and scientific application.
- Learnaboutbiotechnologyanditsconceptsaswell asvariousscopesin Biotechnology.

Practical Syllabus

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ExercisesinCellBiology:

1. PrincipleandusesofMicroscopy.
2. Squashpreparationforthestudyofmitosisintheonionroottip,permanentslidesof mitosis (all stages).
3. Squashpreparationforthestudyofmeiosisingrasshopperorcockroachtestes, permanent slides of meiosis (all stages).
4. Studyofgiantchromosomesinsalivaryglandsof*Chironomus*or*Drosophila*larva.
5. Preparationofbloodsmearanddifferentialstainingofbloodcells.

ExercisesinGenetics:

6. Lifecycle of *Drosophila*andanideaaboutitsculture.
7. Identificationofmaleandfemale *Drosophila*.
8. Identificationofwildandmutant(yellowbody,ebony,vestigialwingandwhiteeye).
9. Studyofpermanentpreparedslides:Sexcombandsalivaryglandchromosomes.
10. Numerical exercisesonMonohybridanddihybridcross.

ExercisesinBiotechnology:

11. StudyofLabinstruments:Centrifuge, Autoclave,pHmeter.

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12. IsolationofDNAfromcheekcells.

13. SeparationofDNAby Agarosegelelectrophoresis.

14. MBRT test for milk quality analysis

Scheme of Practical Examination and Distribution of Marks

S.No.	Practical Exercises	Regular Students	Ex. /N.C. Students
1.	Exercise in Cell Biology	**8**	**12**
2.	Exercise in Genetics	**7**	**6**
3.	Exercise in Biotechnology	**7**	**6**
4.	Identification and comments on Spots (1 to 8)	**16**	**16**
5.	Viva Voce	**7**	**10**
6.	Class Record and report	**5**	
		50	**50**

Notes:

***Internalmarksforregularstudentsonly**

1. With reference to microscopic slides, in case of non-availability, the exercise should be substituted with diagrams / photographs.

2. Candidates must keep a record of all work done in the practical class and submit the same for inspection at the time of the practical examination.

3. Mounting material for permanent preparations would be as per the syllabus or as available through collection and culture methods.

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